

Atropine (systemic): Drug information - UpToDate

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For additional information see "Atropine (systemic): Patient drug information" and "Atropine (systemic): Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Brand Names: US

- AtroPen [DSC]

Pharmacologic Category

- Anticholinergic Agent;
- Antidote;
- Antispasmodic Agent, Gastrointestinal

Dosing: Adult

Dosage guidance:

Safety: IV doses ≤ 0.25 mg have been associated with paradoxical bradycardia (Ref).

Atrioventricular block, symptomatic and/or hemodynamically unstable

Atrioventricular block, symptomatic and/or hemodynamically unstable (off-label use):

Note: Not recommended for atrioventricular block below the atrioventricular node; sinus rate may increase but atrioventricular conduction may worsen. Only consider using in patients with second-degree or third-degree atrioventricular block occurring at the atrioventricular nodal level as a temporizing measure until a permanent pacemaker can be placed, if clinically indicated. In most cases, pharmacologic agents will not be effective and temporary transcutaneous or transvenous pacing is necessary as a bridge to permanent pacemaker placement. Atropine may not be effective and may cause paradoxical heart block or sinus arrest in heart transplant recipients who remain denervated (Ref).

IV, IM: 1 mg every 3 to 5 minutes as needed until symptoms resolve or heart rate is stabilized; maximum total dose: 3 mg (Ref).

Bradycardia during neuromuscular blockade reversal

Bradycardia during neuromuscular blockade reversal: IV: 5 to 7 mcg/kg when administered with edrophonium or 15 to 20 mcg/kg when administered with neostigmine (Ref).

Bradycardia, symptomatic and/or hemodynamically unstable

Bradycardia, symptomatic and/or hemodynamically unstable:

Note: Atropine may not be effective and may cause paradoxical heart block or sinus arrest in heart transplant recipients who remain denervated. Atropine may not be effective for type II second-degree or third-degree heart block (Ref).

IV, IM, Intraosseous : 1 mg every 3 to 5 minutes as needed until symptoms resolve or heart rate is stabilized; maximum total dose: 3 mg (Ref).

Endotracheal (least preferred route): 2 to 2.5 mg every 3 to 5 minutes (Ref).

Inhibit salivation and secretions

Inhibit salivation and secretions (preanesthesia): IM, IV, SUBQ: 0.5 to 1 mg 30 to 60 minutes before procedure; repeat every 4 to 6 hours as needed; maximum total dose: 3 mg.

Muscarine-containing mushroom poisoning

Muscarine-containing mushroom poisoning (off-label dose): IV: 1 to 2 mg; titrate and repeat as needed to reverse symptoms (ie, titrate to achieve decreased bronchial secretions) (Ref).

Organophosphate or carbamate insecticide or nerve agent poisoning

Organophosphate or carbamate insecticide or nerve agent poisoning:

Note: For the most contemporary approach to patient selection and dosing recommendations, consultation with a poison center or a toxicologist is highly recommended. Atropine dosing varies considerably with the severity and type of poisoning. The total amount of atropine used for carbamate poisoning is usually less than with organophosphate insecticide or nerve agent poisoning. Severely poisoned patients require total atropine doses of hundreds of milligrams over the course of days (Ref). Administer atropine via continuous IV infusion in patients requiring large doses (Ref). For organophosphate insecticide and nerve agent toxicity, use in combination with pralidoxime. Available routes of administration include IV (preferred) and IM; other available routes include intraosseous (off label).

IV, IM, Intraosseous (off label):

Initial bolus dose:

Mild symptoms (eg, miosis, severe rhinorrhea): **IV, IM, Intraosseous:** 1 to 2 mg (Ref).

Moderate to severe symptoms (eg, respiratory distress, vomiting, seizures, bronchorrhea, bronchospasm, bradycardia): **IV (preferred), IM, Intraosseous:** 2 to 5 mg (Ref).

Repeat bolus dose(s): **IV (preferred), IM, Intraosseous:** Double the previous dose every 3 to 5 minutes until pulmonary symptoms are improved (clearing of respiratory secretions and the cessation of bronchoconstriction/bronchospasm) (Ref).

Maintenance: **Note:** For patients with moderate or severe symptoms, start an IV continuous infusion after the desired response is achieved with bolus dosing (Ref).

IV: Administer 10% to 20% of the total cumulative bolus dose as an IV continuous infusion per hour (eg, if 18 mg given by bolus is required to achieve the desired response, start an IV continuous infusion of 1.8 to 3.6 mg per hour); titrate to clinical effect and taper as the patient recovers (Ref).

IM (auto-injector):

Note: The only available product is the Rafa Atropine Auto-Injector approved under an Emergency Use Authorization (EUA) for the US stockpile (Ref). It is not available for sale to citizens. The Auto-Injector is authorized for the initial treatment of muscarinic symptoms of known or suspected poisoning in individuals exposed to nerve agents or certain insecticides (organophosphorus and/or carbamate).

Mild symptoms (miosis and severe rhinorrhea): Administer 2 mg (1 green-label injector) as soon as an exposure is known or strongly suspected. If severe symptoms develop after the first dose, 2 additional doses should be repeated in rapid succession 10 minutes after the first dose; do not administer more than 3 doses. If profound anticholinergic effects occur in the absence of excessive bronchial secretions, further doses of atropine should be withheld (Ref).

Moderate to severe symptoms (≥ 1 severe symptoms): Immediately administer **three** 2 mg green-label injectors in rapid succession (total dose: 6 mg) (Ref).

Stress echocardiography

Stress echocardiography (adjunctive agent) (diagnostic agent) (off-label use): IV: 0.25 to 0.5 mg every 1 minute as needed until 85% of age-predicted maximum heart rate is achieved; maximum total dose: 1 to 2 mg (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling (has not been studied).

Dosing: Liver Impairment: Adult

There are no dosage adjustments provided in the manufacturer's labeling (has not been studied).

Dosing: Older Adult

Refer to adult dosing.

Dosing: Pediatric

(For additional information see "Atropine (systemic): Pediatric drug information")

Dosage guidance:

Dosage form information: Atropen auto-injector has been discontinued in the United States for >1 year.

Bradycardia

Bradycardia:

Infants, Children, and Adolescents:

IV, Intraosseous: 0.02 mg/kg/dose; minimum dose: 0.1 mg/dose, maximum dose: 0.5 mg/dose; may repeat once in 5 minutes; reserve use for those patients unresponsive to improved oxygenation and epinephrine (Ref); some have suggested the minimum dose should not be used in patients <5 kg (Ref).

Endotracheal: 0.04 to 0.06 mg/kg/dose; may repeat once if needed (Ref).

Inhibit salivation and secretions

Inhibit salivation and secretions (preoperative/intraoperative):

Infants and Children <12 years: IM, IV, SUBQ: 0.02 mg/kg/dose; maximum dose: 0.5 mg/dose; administer first dose 30 to 60 minutes preoperatively and then repeat every 4 to 6 hours as needed; maximum **total** dose: 1 mg/procedure.

Children ≥ 12 years and Adolescents: IM, IV, SUBQ: 0.02 mg/kg/dose; maximum dose: 1 mg/dose; administer first dose 30 to 60 minutes preoperatively and then repeat every 4 to 6 hours as needed; maximum **total** dose: 2 mg/procedure.

Intubation; emergent

Intubation; emergent (premedication): Note: Routine use not recommended for preintubation in infants and children; atropine may be considered in situations with a high risk of bradycardia (eg, succinylcholine use) (Ref) or septic shock (Ref).

Infants, Children, and Adolescents: IV, Intraosseous: Usual dose: 0.02 mg/kg/dose with no minimum dose; recommended range: 0.01 to 0.02 mg/kg/dose; maximum dose: 0.5 mg/dose (Ref).

Muscarine-containing mushroom poisoning

Muscarine-containing mushroom poisoning: Limited data available: Infants, Children, and Adolescents: IV: 0.02 mg/kg/dose; minimum dose: 0.1 mg. Titrate and repeat as needed to reverse symptoms (ie, titrate to achieve decreased bronchial secretions) (Ref).

Organophosphate or carbamate insecticide or nerve agent poisoning

Organophosphate or carbamate insecticide or nerve agent poisoning:

Note: If exposure is known or suspected, antidotal therapy should be given as soon as symptoms appear; do not wait for confirmation. The dose of atropine required varies considerably with the severity of poisoning. The total amount of atropine used for carbamate poisoning is usually less than with organophosphate insecticide or nerve agent poisoning. Severely poisoned patients may exhibit significant tolerance to atropine; ≥ 2 times the suggested doses may be needed. Titrate to pulmonary status (decreased bronchial secretions); consider administration of atropine via continuous IV infusion in patients requiring large doses of atropine (Ref). Once patient is stable for a period of time, the dose/dosing frequency may be decreased. Pralidoxime is a component of the management of organophosphate insecticide and nerve agent toxicity; refer to pralidoxime for the specific route and dose.

IV, IM, Intraosseous (Ref):

Infants and Children: Initial: 0.05 to 0.1 mg/kg; repeat every 3 to 5 minutes as needed, double the dose if previous dose does not induce atropinization (Ref). Maintain atropinization by administering repeat doses as needed for ≥ 2 to 12 hours based on recurrence of symptoms (Ref).

Adolescents: Initial: 1 to 3 mg; repeat every 3 to 5 minutes as needed, doubling the dose if previous dose does not induce atropinization. Maintain atropinization by administering repeat doses as needed for ≥ 2 to 12 hours based on recurrence of symptoms (Ref).

Continuous IV infusion: Infants, Children, and Adolescents: Following atropinization, administer 10% to 20% of the total loading dose required to induce atropinization as a continuous IV infusion per hour; adjust as needed to maintain adequate atropinization without atropine toxicity (Ref).

IM (AtroPen): Infants, Children, and Adolescents: Number of doses dependent upon symptom severity:

Weight-directed dosing:

<7 kg (<15 lb): 0.25 mg/dose (yellow pen).

7 to 18 kg (15 to 40 lb): 0.5 mg/dose (blue pen).

>18 to 41 kg (>40 to 90 lb): 1 mg/dose (dark red pen).

>41 kg (>90 lb): 2 mg/dose (green pen).

Mild symptoms (≥ 2 mild symptoms): Administer the weight-directed dose listed above as soon as an exposure is known or strongly suspected. If severe symptoms develop after the first dose, 2 additional doses should be repeated in rapid succession 10 minutes after the first dose; do not administer more than 3 doses. If profound anticholinergic effects occur in the absence of excessive bronchial secretions, further doses of atropine should be withheld. Mild symptoms of insecticide or nerve agent poisoning, as provided by manufacturer in the AtroPen product labeling to guide therapy, include: Blurred vision, bradycardia, breathing difficulties, chest tightness, coughing, drooling, miosis, muscular twitching, nausea, runny nose, salivation increased, stomach cramps, tachycardia, teary eyes, tremor, vomiting, or wheezing.

Severe symptoms (≥ 1 severe symptom): Immediately administer **three** weight-directed doses in rapid succession (Ref). Severe symptoms of insecticide or nerve agent poisoning, as provided by manufacturer in the AtroPen product labeling to guide therapy, include: Breathing difficulties (severe), confused/strange behavior, defecation (involuntary), muscular twitching/generalized weakness (severe), respiratory secretions (severe), seizure, unconsciousness, urination (involuntary); **Note:** Infants may become drowsy or unconscious with muscle floppiness as opposed to muscle twitching.

Endotracheal: Infants, Children, and Adolescents: Increase the dose by 2 to 3 times the usual IV dose. Mix with 3 to 5 mL of normal saline and administer. Flush with 3 to 5 mL of NS and follow with 5 assisted manual ventilations (Ref).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling.

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified. Frequency not always defined. Severity and frequency of adverse reactions are dose related.

Cardiovascular: Asystole, atrial arrhythmia, atrial fibrillation, atrioventricular dissociation (transient), bigeminy, bradycardia, chest pain, decreased blood pressure, ECG changes (prolonged P wave, shortened PR segment, R on T phenomenon, shortened RT duration, prolonged QT interval, widening of QRS Complex, flattened T wave, repolarization abnormalities, ST segment elevation, retrograde conduction), ectopic beats (atrial), extrasystoles (nodal, ventricular, supraventricular), flushing, increased blood pressure, left heart failure, myocardial infarction, nodal arrhythmia (no P wave on ECG), palpitations, sinus tachycardia, supraventricular tachycardia (including junctional tachycardia), tachycardia, trigeminy, ventricular arrhythmia (including flutter), ventricular fibrillation, ventricular flutter, ventricular premature contractions, ventricular tachycardia, weak pulse (or impalpable peripheral pulses)

Central nervous system: Abnormal electroencephalogram (runs of alpha waves, increase in photic stimulation, and signs of drowsiness), agitation (children), amnesia, anxiety, ataxia, behavioral changes, coma, confusion, decreased deep tendon reflex, delirium, dizziness, drowsiness, dysarthria, dysmetria, emotional disturbance, excitement, feeling hot, hallucination (visual or aural), headache, hyperpyrexia, hyperreflexia, hypertonia, insomnia, intoxicated feeling, irritability (children), lack of concentration, lethargy (children), mania, myoclonus, neurologic abnormality, nocturnal enuresis, opisthotonus, paranoia, positive Babinski sign, restlessness, seizure (generally tonic-clonic), stupor, vertigo

Dermatologic: Anhidrosis, cold skin, dermatitis, dry and hot skin, erythematous rash, hyperhidrosis, macular eruption, maculopapular rash, papular rash, scarlatiniform rash, skin rash

Endocrine & metabolic: Dehydration, hyperglycemia, hypoglycemia, hypokalemia, hyponatremia, increased thirst, loss of libido

Gastrointestinal: Abdominal and bladder distension, abdominal pain, constipation, delayed gastric emptying, diminished bowel sounds, dry mucous membranes, dysphagia, malabsorption, nausea, oral lesion, paralytic ileus, salivation, vomiting, xerostomia

Genitourinary: Difficulty in micturition, impotence, urinary hesitancy, urinary retention, urinary urgency

Hematologic & oncologic: Abnormal erythrocytes (increased), decreased hemoglobin, increased hemoglobin, leukocytosis, petechiae

Hypersensitivity: Hypersensitivity reaction

Local: Injection site reaction

Neuromuscular & skeletal: Laryngospasm, muscle twitching, weakness

Ophthalmic: Abnormal eye movements (cyclophoria and heterophoria), angle-closure glaucoma (acute), blepharitis, blindness, blurred vision, conjunctivitis, crusted eyelid, cycloplegia, decreased

accommodation, decreased visual acuity, dry eye syndrome, eye irritation, lacrimation, mydriasis, photophobia, strabismus

Renal: Increased blood urea nitrogen

Respiratory: Bradypnea, changes in respiration (labored respiration), cyanosis, dyspnea, laryngitis, pulmonary edema, respiratory failure, stridor (inspiratory), tachypnea

Miscellaneous: Failure to thrive, fever (secondary to decreased sweat gland activity), swelling (children)

Contraindications

There are no contraindications listed in the manufacturer's US labeling.

Canadian labeling: Hypersensitivity to atropine or any component of the formulation; GI obstruction, glaucoma (known or suspected), pyloric stenosis or prostatic hypertrophy (except in doses typically used for preanesthesia).

Warnings/Precautions

Concerns related to adverse effects:

- Hypersensitivity: Hypersensitivity reactions, including anaphylactic reactions, may occur.
- Hyperthermia: Atropine may inhibit sweating and possibly lead to heat-related injury or hyperthermia in patients exposed to warm environments or exercise.
- Psychosis: Can occur in sensitive individuals or following use of excessive doses.

Disease-related concerns:

- Arrhythmias: Avoid relying on atropine for effective treatment of type II second-degree or third-degree AV block (with or without a new wide QRS complex). Asystole or bradycardic PEA: Although no evidence exists for significant detrimental effects, routine use is unlikely to have a therapeutic benefit and is no longer recommended (ACLS 2010).
- Cardiovascular disease: Use with caution in patients with myocardial ischemia, heart failure, tachyarrhythmias (including sinus tachycardia), and/or hypertension; treatment-related blood pressure increases and tachycardia may lead to ischemia, precipitate an MI, or increase arrhythmogenic potential.
- Chronic lung disease: Use with caution in patients with chronic lung disease; may cause thickening of bronchial secretions and formation of dangerous viscid plugs.
- Glaucoma: Use with caution in patients with severe narrow-angle glaucoma; may precipitate acute glaucoma.
- Hepatic impairment: Use with caution in patients with hepatic impairment; effects of atropine may be prolonged in severe hepatic impairment.
- Hiatal hernia: Use with caution in patients with hiatal hernia associated with reflux esophagitis.
- Hyperthyroidism: Use with caution in patients with hyperthyroidism.
- Myasthenia gravis: Use with extreme caution when used to treat side effects of acetylcholinesterase inhibition or avoid; may precipitate a myasthenic crisis.
- Neuropathy: Use with caution in patients with autonomic neuropathy.
- Pyloric stenosis: Use with caution in patients with partial pyloric stenosis; may cause complete pyloric obstruction.
- Renal impairment: Use with caution in patients with renal impairment; effects of atropine may be prolonged in severe renal impairment.
- Urinary retention: Use with caution in patients with urinary obstruction; may cause urinary retention.

Special populations:

- Heart transplant recipients: Atropine will likely be ineffective in treatment of bradycardia due to lack of vagal innervation of the transplanted heart. Cholinergic reinnervation may occur over time (years), so atropine may be used cautiously; however, some may experience paradoxical slowing of the heart rate and high-degree AV block upon administration (ACLS 2010; Bernheim 2004).
- Pediatric: Children may be more sensitive to the anticholinergic effects of atropine. Use with caution in children with spastic paralysis.

Dosage form specific issues:

- Benzyl alcohol and derivatives: Some dosage forms may contain benzyl alcohol and/or sodium benzoate/benzoic acid; benzoic acid (benzoate) is a metabolite of benzyl alcohol; large amounts of benzyl alcohol (≥ 99 mg/kg/day) have been associated with a potentially fatal toxicity ("gasping syndrome") in neonates; the "gasping syndrome" consists of metabolic acidosis, respiratory distress, gasping respirations, CNS dysfunction (including convulsions, intracranial hemorrhage), hypotension, and cardiovascular collapse (AAP ["Inactive" 1997]; CDC 1982); some data suggest that benzoate displaces bilirubin from protein binding sites (Ahlfors 2001); avoid or use dosage forms containing benzyl alcohol and/or benzyl alcohol derivative with caution in neonates. See manufacturer's labeling.

Other warnings/precautions:

- Appropriate use: Anticholinesterase poisoning: Refer to Lexidrug Toxicology: Organophosphates.

Warnings: Additional Pediatric Considerations

Several reports in the literature have described neonates developing central anticholinergic syndrome after receiving atropine for bradycardia at doses of 0.1 mg, the minimum dose that has been described for preventing paradoxical bradycardia. When evaluated based on weight, this dose exceeded 0.02 mg/kg in all cases (Gillick 1974; Prakash 2017; Rizzi 2004); some literature has recommended against using a minimum dose in patients < 5 kg to prevent overdose (Barrington 2011; Prakash 2017).

Product Availability

Atropen auto-injector has been discontinued in the United States for > 1 year. The only available product is the Rafa Atropine Auto-Injector approved under an Emergency Use Authorization for the US stockpile. It is not available for sale to citizens.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Solution, Injection, as sulfate:

Generic: 8 mg/20 mL (20 mL)

Solution, Injection, as sulfate [preservative free]:

Generic: 0.4 mg/mL (1 mL [DSC]); 1 mg/mL (1 mL [DSC])

Solution, Intravenous, as sulfate [preservative free]:

Generic: 0.4 mg/mL (1 mL); 1 mg/mL (1 mL)

Solution Auto-injector, Intramuscular, as sulfate:

AtroPen: 0.5 mg/0.7 mL (0.7 mL [DSC]); 1 mg/0.7 mL (0.7 mL [DSC]); 2 mg/0.7 mL (0.7 mL [DSC]) [pyrogen free; contains phenol]

Generic: 2 mg/0.7 mL (0.7 mL)

Solution Auto-injector, Intramuscular, as sulfate [preservative free]:

AtroPen: 0.25 mg/0.3 mL (0.3 mL [DSC]) [pyrogen free]

Solution Prefilled Syringe, Injection, as sulfate:

Generic: 1 mg/10 mL (10 mL)

Solution Prefilled Syringe, Injection, as sulfate [preservative free]:

Generic: 0.25 mg/5 mL (5 mL); 0.5 mg/5 mL (5 mL); 1 mg/10 mL (10 mL)

Generic Equivalent Available: US

Yes

Pricing: US

Solution (Atropine Sulfate Injection)

8 mg/20 mL (per mL): \$2.11 - \$4.58

Solution (Atropine Sulfate Intravenous)

0.4 mg/mL (per mL): \$12.77 - \$14.50

1 mg/mL (per mL): \$17.76 - \$20.71

Solution Auto-injector (Atropine Sulfate Intramuscular)

2 mg/0.7 mL (per 0.7 mL): \$240.00

Solution Prefilled Syringe (Atropine Sulfate Injection)

0.25 mg/5 mL (per mL): \$7.87 - \$7.88

0.5 mg/5 mL (per mL): \$2.29

1 mg/10 mL (per mL): \$1.20 - \$1.37

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Solution, Injection:

Generic: 0.6 mg/mL (1 mL)

Solution, Injection, as sulfate:

Generic: 0.1 mg/mL ([DSC]); 0.4 mg/mL (1 mL, 10 mL)

Solution Prefilled Syringe, Injection:

Generic: 0.5 mg/5 mL (5 mL); 1 mg/5 mL (5 mL)

Prescribing and Access Restrictions

The auto-injector formulation is available for use primarily by the Department of Defense.

Administration: Adult

IM: Auto-injector: Administer to the outer thigh. Firmly grasp the auto-injector with the green tip (0.5 mg, 1 mg, and 2 mg auto-injector) pointed down; remove the yellow safety release (0.5 mg, 1 mg, and 2 mg auto-injector). Jab the green tip at a 90° angle against the outer thigh; may be administered through clothing as long as pockets at the injection site are empty. In thin patients, bunch up the thigh prior to injection. Hold the auto-injector in place for 10 seconds following the injection; remove the auto-injector and massage the injection site. After administration, the needle will be visible; if the needle is not visible, repeat the above steps. After use, bend the needle against a hard surface (needle does not retract) to avoid accidental injury.

IV: Administer undiluted by rapid IV injection; slow injection may result in paradoxical bradycardia. In bradycardia, atropine administration should not delay treatment with external pacing.

Intraosseous: May administer intraosseous if needed.

Endotracheal: Dilute in ≤ 10 mL NS or sterile water. Absorption may be greater with sterile water. Stop compressions (if using for cardiac arrest), spray the drug quickly down the tube. Follow immediately with several quick insufflations and continue chest compressions.

SUBQ: May administer subcutaneous if needed.

Preparation for Administration: Adult

Parenteral: **Mass casualty event:** Preparation of bulk atropine solution: Add atropine sulfate powder to 100 mL of NS in polyvinyl chloride bags to yield a final concentration of 1 mg/mL (Ref).

Administration: Pediatric

Endotracheal: May administer dose undiluted, followed by flush with 1 to 5 mL of NS after endotracheal administration or may further dilute in NS or sterile water prior to administration (ie, 1 to 2 mg in ≤ 10 mL of NS or sterile water); follow with 5 assisted manual ventilations (Ref). If using for cardiac arrest, stop compressions during drug administration and resume chest compressions following manual ventilation.

Flush volume based on indication:

Bradycardia:

Neonates: Flush with ≥ 1 mL NS (Ref).

Infants, Children, and Adolescents: Flush with 5 mL NS (Ref).

Organophosphate or carbamate insecticide or nerve agent poisoning: Flush with 3 to 5 mL NS (Ref).

Parenteral:

IV: Administer undiluted by rapid IV injection; slow injection may result in paradoxical bradycardia.

IM: AtroPen: Administer to the outer thigh. Firmly grasp the autoinjector with the green tip (0.5 mg, 1 mg, and 2 mg autoinjector) or black tip (0.25 mg autoinjector) pointed down; remove the yellow safety release (0.5 mg, 1 mg, and 2 mg autoinjector) or gray safety release (0.25 autoinjector). Firmly jab the green tip at a 90° angle against the outer thigh; may be administered through clothing as long as pockets at the injection site are empty. In thin patients or patients < 6.8 kg (15 lb), bunch up the thigh prior to injection. Hold the autoinjector in place for 10 seconds following the injection; remove the autoinjector and massage the injection site. After administration, the needle will be visible; if the needle is not visible, repeat the above steps with more pressure. After use, bend the needle against a hard surface (needle does not retract) to avoid accidental injury.

Preparation for Administration: Pediatric

Parenteral: **Mass chemical terrorism:** Preparation of bulk atropine solution: Add atropine sulfate powder to 100 mL NS in polyvinyl chloride bags to yield a final concentration of 1 mg/mL (Ref).

Storage/Stability

Store injection at room temperature; avoid freezing. After initial use of multiple-dose vial, discard within 24 hours. In addition, AtroPen should be protected from light. Preparation of bulk atropine

solution for mass chemical terrorism at a concentration of 1 mg/mL is stable for 72 hours at 4°C to 8°C (39°F to 46°F); 20°C to 25°C (68°F to 77°F); 32°C to 36°C (90°F to 97°F) (Dix 2003).

Pharmacy supply of emergency antidotes:

Guidelines suggest that at least 45 to 165 mg be stocked. Suggested amount is stated to be a sufficient quantity to treat 1 patient weighing 100 kg for an initial 8- to 24-hour period; actual amount to stock should take into account site-specific and population-specific needs. Hospitals serving agricultural regions where organophosphate or carbamate insecticides are used should ensure that they maintain adequate supplies (Dart 2018).

In severe poisonings, the amount needed may exceed the standard recommendation (Dart 2009).

Mass casualty event secondary to an acetylcholinesterase inhibitor: In the unlikely occurrence of a mass casualty event secondary to an acetylcholinesterase inhibitor, supplies of atropine may rapidly deplete, including predeployed CHEMPACK containers. The US Food and Drug Administration (FDA) no longer recommends the use of expired medications, including auto-injectors (Ref). Consider stocking atropine as a ≥ 5 g bulk pharmaceutical grade powder in order to prepare large quantities of atropine solution for administration after an act of terrorism involving a nerve agent (Ref).

Use: Labeled Indications

Bradycardia during neuromuscular blockade reversal: Adjuvant use during neuromuscular blockade reversal with anticholinesterases (eg, edrophonium, neostigmine) to manage bradycardia.

Bradycardia, symptomatic: Treatment of symptomatic sinus bradycardia.

Inhibition of salivation and secretions (preanesthesia): Preoperative/preanesthetic medication to inhibit salivation and secretions.

Muscarine-containing mushroom poisoning: Treatment of symptoms from muscarine-containing mushroom poisoning.

Organophosphate or carbamate insecticide or nerve agent poisoning: Antidote for anticholinesterase poisoning (carbamate insecticides, nerve agents, organophosphate insecticides).

Use: Off-Label: Adult

Atrioventricular block, symptomatic and/or hemodynamically unstable; Stress echocardiography (adjunctive agent) (diagnostic agent)

Medication Safety Issues

Older Adult: High-Risk Medication:

Beers Criteria: Atropine is identified in the Beers Criteria as a potentially inappropriate medication to be avoided in patients 65 years and older (independent of diagnosis or condition) due to its highly anticholinergic properties and uncertain effectiveness as an antispasmodic (Beers Criteria [AGS 2023]).

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Acetylcholinesterase Inhibitors: May decrease therapeutic effects of Agents with Clinically Relevant Anticholinergic Effects. Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Acetylcholinesterase Inhibitors. *Risk C: Monitor*

Acridinium: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk X: Avoid*

Acrivastine: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Alpha1-Agonists: Atropine (Systemic) may increase hypertensive effects of Alpha1-Agonists. *Risk C: Monitor*

Amantadine: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Amezinium: Atropine (Systemic) may increase stimulatory effects of Amezinium. *Risk C: Monitor*

Benperidol: Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Benperidol. *Risk C: Monitor*

Benztropine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Benztropine. *Risk C: Monitor*

Beta-Acetyldigoxin: Atropine (Systemic) may increase serum concentrations of Beta-Acetyldigoxin. *Risk C: Monitor*

Biperiden: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Biperiden. *Risk C: Monitor*

Bornaprine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Bornaprine. *Risk C: Monitor*

Botulinum Toxin-Containing Products: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Bromperidol: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Bromperidol. *Risk C: Monitor*

Bucizine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Bucizine. *Risk C: Monitor*

Cannabinoid-Containing Products: Agents with Clinically Relevant Anticholinergic Effects may increase tachycardic effects of Cannabinoid-Containing Products. *Risk C: Monitor*

Chlorprothixene: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Chlorprothixene. *Risk C: Monitor*

Cimetropium: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Cimetropium. *Risk X: Avoid*

CloZAPine: Agents with Clinically Relevant Anticholinergic Effects may increase constipating effects of CloZAPine. Management: Consider alternatives to this combination whenever possible. If combined, monitor closely for signs and symptoms of gastrointestinal hypomotility and consider prophylactic laxative treatment. *Risk D: Consider Therapy Modification*

Cyclizine: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Darifenacin: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Darifenacin. *Risk C: Monitor*

Dicyclomine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Dicyclomine. *Risk C: Monitor*

Dimethindene (Systemic): Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Dimethindene (Systemic). *Risk C: Monitor*

DroNABinol: Agents with Clinically Relevant Anticholinergic Effects may increase tachycardic effects of DroNABinol. *Risk X: Avoid*

Eluxadoline: Agents with Clinically Relevant Anticholinergic Effects may increase constipating effects of Eluxadoline. *Risk X: Avoid*

EPHEDrine (Systemic): Atropine (Systemic) may increase therapeutic effects of EPHEDrine (Systemic). *Risk C: Monitor*

Fesoterodine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Fesoterodine. *Risk C: Monitor*

FluPHENAZine: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Gastrointestinal Agents (Prokinetic): Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Gastrointestinal Agents (Prokinetic). *Risk C: Monitor*

Gepotidacin: May decrease anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Glucagon: Agents with Clinically Relevant Anticholinergic Effects may increase adverse/toxic effects of Glucagon. Specifically, the risk of gastrointestinal adverse effects may be increased. *Risk C: Monitor*

Glycopyrrolate (Oral Inhalation): Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Glycopyrrolate (Oral Inhalation). *Risk X: Avoid*

Glycopyrrolate (Systemic): Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Glycopyrrolate (Systemic). *Risk C: Monitor*

Glycopyrronium (Topical): May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk X: Avoid*

Haloperidol: May increase adverse/toxic effects of Agents with Clinically Relevant Anticholinergic Effects. Specifically, intraocular pressure may be increased. *Risk C: Monitor*

Hyoscyamine: Anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects and Hyoscyamine may increase with coadministration. *Risk C: Monitor*

Ipratropium (Nasal): May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Ipratropium (Oral Inhalation): May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk X: Avoid*

Itopride: Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Itopride. *Risk C: Monitor*

Levosulpiride: Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Levosulpiride. *Risk X: Avoid*

Macimorelin: Coadministration of Atropine (Systemic) and Macimorelin may alter diagnostic results. *Risk X: Avoid*

Melperone: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Methotrimeprazine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Methotrimeprazine. *Risk C: Monitor*

Methscopolamine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Methscopolamine. *Risk C: Monitor*

Mirabegron: Agents with Clinically Relevant Anticholinergic Effects may increase adverse/toxic effects of Mirabegron. *Risk C: Monitor*

Nitroglycerin: Agents with Clinically Relevant Anticholinergic Effects may decrease absorption of Nitroglycerin. Specifically, anticholinergic agents may decrease the dissolution of sublingual nitroglycerin tablets, possibly impairing or slowing nitroglycerin absorption. *Risk C: Monitor*

OLANzapine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of OLANzapine. *Risk C: Monitor*

Opioid Agonists: Agents with Clinically Relevant Anticholinergic Effects may increase adverse/toxic effects of Opioid Agonists. Specifically, the risk for constipation and urinary retention may be increased with this combination. *Risk C: Monitor*

Opipramol: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Oxatomide: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk X: Avoid*

OxyBUTYnin: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of OxyBUTYnin. *Risk C: Monitor*

Perazine: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Periciazine: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Perphenazine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Perphenazine. *Risk C: Monitor*

Potassium Chloride: Agents with Clinically Relevant Anticholinergic Effects may increase ulcerogenic effects of Potassium Chloride. Management: Patients on drugs with substantial anticholinergic effects should avoid using any solid oral dosage form of potassium chloride. *Risk X: Avoid*

Potassium Citrate: Agents with Clinically Relevant Anticholinergic Effects may increase ulcerogenic effects of Potassium Citrate. Management: Patients on drugs with substantial anticholinergic effects should avoid using any solid oral dosage form of potassium citrate. *Risk X: Avoid*

Pramlintide: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. These effects are specific to the GI tract. *Risk X: Avoid*

Promethazine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Promethazine. *Risk C: Monitor*

Propantheline: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Propantheline. *Risk C: Monitor*

Propiverine: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

QuiNIDine: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk C: Monitor*

Ramosetron: Agents with Clinically Relevant Anticholinergic Effects may increase constipating effects of Ramosetron. *Risk C: Monitor*

Revefenacin: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Revefenacin. *Risk X: Avoid*

Ritodrine: Atropine (Systemic) may increase adverse/toxic effects of Ritodrine. *Risk C: Monitor*

Rivastigmine: Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Rivastigmine. Rivastigmine may decrease therapeutic effects of Agents with Clinically Relevant Anticholinergic Effects. Management: Use of rivastigmine with an anticholinergic agent is not recommended unless clinically necessary. If the combination is necessary, monitor for reduced anticholinergic effects. *Risk D: Consider Therapy Modification*

Scopolamine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Scopolamine. *Risk C: Monitor*

Secretin: Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Secretin. Management: Avoid concomitant use of anticholinergic agents and secretin. Discontinue anticholinergic agents at least 5 half-lives prior to administration of secretin. *Risk D: Consider Therapy Modification*

Sincalide: Drugs that Affect Gallbladder Function may decrease therapeutic effects of Sincalide. Management: Consider discontinuing drugs that may affect gallbladder motility prior to the use of sincalide to stimulate gallbladder contraction. *Risk D: Consider Therapy Modification*

Sofpironium: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Sofpironium. *Risk X: Avoid*

Thiazide and Thiazide-Like Diuretics: Agents with Clinically Relevant Anticholinergic Effects may increase serum concentrations of Thiazide and Thiazide-Like Diuretics. *Risk C: Monitor*

Thiothixene: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Thiothixene. *Risk C: Monitor*

Tiaprside: Agents with Clinically Relevant Anticholinergic Effects may decrease therapeutic effects of Tiaprside. *Risk C: Monitor*

Tiotropium: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Tiotropium. *Risk X: Avoid*

Tolterodine: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Tolterodine. *Risk C: Monitor*

Topiramate: Agents with Clinically Relevant Anticholinergic Effects may increase adverse/toxic effects of Topiramate. *Risk C: Monitor*

Tricyclic Antidepressants: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Tricyclic Antidepressants. *Risk C: Monitor*

Trimethobenzamide: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Trimethobenzamide. *Risk C: Monitor*

Trospium: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Trospium. *Risk C: Monitor*

Umeclidinium: May increase anticholinergic effects of Agents with Clinically Relevant Anticholinergic Effects. *Risk X: Avoid*

Zuclopenthixol: Agents with Clinically Relevant Anticholinergic Effects may increase anticholinergic effects of Zuclopenthixol. *Risk C: Monitor*

Pregnancy Considerations

Atropine crosses the placenta.

In general, medications used as antidotes should take into consideration the health and prognosis of the mother; antidotes should be administered to pregnant women if there is a clear indication for use and should not be withheld because of fears of teratogenicity (Bailey 2003). Medications used for the treatment of cardiac arrest in pregnancy are the same as in the non-pregnant woman. Doses and indications should follow current Advanced Cardiovascular Life Support guidelines. Appropriate medications should not be withheld due to concerns of fetal teratogenicity (Jeejeebhoy [AHA] 2015).

Breastfeeding Considerations

Atropine is present in breast milk.

According to the manufacturer, the decision to breastfeed during therapy should consider the risk of infant exposure, the benefits of breastfeeding to the infant, and the benefits of treatment to the mother.

Monitoring Parameters

Heart rate, blood pressure, pulse, mental status; intravenous administration requires a cardiac monitor

Organophosphate or carbamate insecticide or nerve agent poisoning: Heart rate, blood pressure, respiratory status, oxygenation secretions. Maintain atropinization with repeated dosing as indicated by clinical status. Crackles in lung bases, or continuation of cholinergic signs, may be signs of inadequate dosing. Pulmonary improvement may not parallel other signs of atropinization. Monitor for signs and symptoms of atropine toxicity (eg, fever, muscle fasciculations, delirium); if toxicity occurs, discontinue atropine and monitor closely.

Consult individual institutional policies and procedures.

Mechanism of Action

Blocks the action of acetylcholine at parasympathetic sites in smooth muscle, secretory glands, and the CNS; increases cardiac output, dries secretions. Atropine reverses the muscarinic effects of cholinergic poisoning due to agents with acetylcholinesterase inhibitor activity by acting as a competitive antagonist of acetylcholine at muscarinic receptors. The primary goal in cholinergic poisonings is reversal of bronchorrhea and bronchoconstriction. Atropine has no effect on the nicotinic receptors responsible for muscle weakness, fasciculations, and paralysis; concurrent administration of pralidoxime is necessary to reverse the nicotinic effects associated with organophosphate insecticide or nerve agent toxicity.

Pharmacokinetics (Adult Data Unless Noted)

Onset of action:

Inhibition of salivation: IM: Within 30 minutes; maximum effect: 30 to 60 minutes (Mirakhur 1980; Volz-Zang 1995)

Increased heart rate:

IM: Within 15 to 30 minutes (Kentala 1990; Volz-Zang 1995); maximum effect: 45 to 60 minutes (Mirakhur 1980; Volz-Zang 1995)

IV: Immediate; maximum effect: 0.7 to 4 minutes (Lonnerholm 1975; Santini 1999)

Duration: Inhibition of salivation: IM: ≤ 4 hours (Mirakhur 1980; Volz-Zang 1995)

Absorption: Rapid and well absorbed from all dosage forms

Distribution: Widely throughout the body; crosses blood-brain barrier

Protein binding: 14% to 44%

Metabolism: Hepatic via enzymatic hydrolysis

Half-life elimination: Children <2 years: 6.9 ± 3 hours; Children >2 years: 2.5 ± 1.2 hours; Adults: 3 ± 0.9 hours; Elderly 65 to 75 years of age: 10 ± 7.3 hours

Time to peak: IM: 30 minutes; IM autoinjector: 3 minutes

Excretion: Urine (13% to 50% as unchanged drug and metabolites)

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Older adult: Elimination half-life is more than doubled.

Sex: AUC and C_{max} are 15% higher in females than males; Elimination half-life is ~ 20 minutes shorter in females than males.

Patient Counseling Points

(For additional information see "Atropine (systemic): Patient drug information")

What is this drug used for?

- It is used to treat some poisonings.

- In surgery, it is used to lower secretions such as saliva.
- It is used to treat muscle spasms of the GI (gastrointestinal) tract, gallbladder system, or urinary system.
- It is used when the heart is not beating.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

- Blurred eyesight
- If bright lights bother your eyes
- Constipation
- Stomach pain
- Upset stomach or throwing up
- Less sweating
- Dizziness or headache
- Dry mouth
- Dry eyes
- Dry nose
- Larger pupils
- Flushing
- Pain where the shot was given

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

- Trouble passing urine
- Chest pain or pressure, a fast heartbeat, or an abnormal heartbeat
- Very bad dizziness or passing out
- Trouble breathing, slow breathing, or shallow breathing
- Feeling confused
- Change in balance
- Hallucinations (seeing or hearing things that are not there)
- Feeling very tired or weak
- Restlessness
- Shakiness
- Swelling of belly
- Severe constipation or stomach pain
- Not able to get or keep an erection
- Lowered interest in sex
- Low mood (depression)
- Change in eyesight, eye pain, or very bad eye irritation

- Not sweating during activities or in warm temperatures
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AR) Argentina: Endotropina | Solucion sulfato de atropina | Sulfato de atropina Norgreen | Sulfato de atropina norgreen forte;
- (AT) Austria: Atropinsulfat Aguettant;
- (AU) Australia: Atropine sulphate;
- (BE) Belgium: Atropine sulfate aguettant;
- (BR) Brazil: Sulfato de atropina;
- (CH) Switzerland: Atropin sulfat | Atropin sulfat aguettant | Atropinium sulfuricum streuli;
- (CN) China: Atropine;
- (CZ) Czech Republic: Atropin | Atropine accord;
- (DE) Germany: Atropin accord | Dysurgal | Min-i-jet atropine;
- (ET) Ethiopia: Atropine sulphate;
- (FI) Finland: Atropin | Atropin aguettant | Atropin stragen | Atropine accord;
- (FR) France: Atropine sulfate aguettant;
- (GB) United Kingdom: Atropine sulphate;
- (HK) Hong Kong: Atropine;
- (HU) Hungary: Atropinum sulfuricum;
- (ID) Indonesia: Atropine;
- (IE) Ireland: Atropine sulfate aguettant | Atropine sulphate;
- (IL) Israel: Atropine;
- (IN) India: Atropine | Atropine sulphate | Atrothem;
- (IT) Italy: Atropina solfato accord | Atropina solfato laboratoire aguettant;
- (JP) Japan: Atquick | Atropine;
- (KE) Kenya: Atropine;
- (KR) Korea, Republic of: Sergal;
- (KW) Kuwait: Atropine sulfate aguettant;
- (LT) Lithuania: Atropin | Dysurgal;
- (LV) Latvia: Atropin;
- (MX) Mexico: Amixteria;
- (MY) Malaysia: Atropine sulphate;
- (NL) Netherlands: Atropine sulf | Atropinesulf | Atropinesulfaat Aguettant;
- (NO) Norway: Atropin | Atropin aguettant | Atropin stragen;

- (NZ) New Zealand: Atropine sulphate;
- (PA) Panama: Atropina sulfato;
- (PK) Pakistan: Atroject | Atropine;
- (PL) Poland: Atropin;
- (PR) Puerto Rico: Sal-tropine;
- (PT) Portugal: Atropina accord | Atropina aguettant | Atropina Labesfal;
- (PY) Paraguay: Atropina sulfato lasca;
- (RO) Romania: Sulfat de atropina accord;
- (RU) Russian Federation: Atropin | Atropine nova;
- (SA) Saudi Arabia: Atropine sulphate;
- (SE) Sweden: Atropin stragen | Atropine accord;
- (SG) Singapore: Atropine | Atropine sulphate;
- (SI) Slovenia: Atropina solfato monico | Atropini sulfas;
- (SK) Slovakia: Atropin;
- (TH) Thailand: Atropine;
- (TW) Taiwan: Atropin sulfate | Atropine | Bellaton | Soreless | Spanton | Spasmatal;
- (UA) Ukraine: Atropin sulfate;
- (UG) Uganda: Atropine sulfate aguettant;
- (UY) Uruguay: Endotropina;
- (ZW) Zimbabwe: Atropine sulphate | Atropine sulphate fresenius | Pharma q atropine

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